Relaxation

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Solution of the practice quiz

1. The decision variables are binary variables: x_i is 1 if a dam is built at location i, and 0 otherwise.

   The objective is to maximize the expected revenues:

   $$\max_{x\in\{0,1\}^4} 16x_1 + 22x_2 + 12x_3 + 8x_4.$$

   Note that we are modeling the problem using 10mCHF as the base unit.

   The budget constraint is given by

   $$5x_1 + 7x_2 + 4x_3 + 3x_4 \leq 14.$$

   Therefore, the integer optimization problem is:

   $$\max_{x\in\{0,1\}^4} 16x_1 + 22x_2 + 12x_3 + 8x_4,$$

   subject to

   $$\begin{aligned} 5x_1 + 7x_2 + 4x_3 + 3x_4 &\leq 14, \\ x_1, x_2, x_3, x_4 &\in \{0, 1\}. \end{aligned}$$

   It is actually a knapsack problem.

2. The relaxation of the problem is

   $$\max_{x\in\{0,1\}^4} 16x_1 + 22x_2 + 12x_3 + 8x_4,$$

subject to

$$\begin{aligned} 5x_1 + 7x_2 + 4x_3 + 3x_4 &\leq 14, \\ x_1, x_2, x_3, x_4 &\geq 0, \\ x_1, x_2, x_3, x_4 &\leq 1. \end{aligned}$$

The optimal solution of the relaxation can be calculated using the simplex algorithm. It is

$$x_1 = 1,\ x_2 = 1,\ x_3 = 0.5,\ x_4 = 0.$$

For the company, this suggests to build dams 1 and 2, and not to build dam 4. Regarding dam 3, building it would not fit the budget. Building dams 1 and 2 will cost 120 mCHF, and the expected benefit is 380 mCHF.

3. We enumerate the feasible solutions. For each of them, we calculate the revenues, the cost, and we check if it is feasible for the budget.

x_1	x_2	x_3	x_4	Revenues	Cost	Feasible?
0	0	0	0	0	0	YES
0	0	0	1	8	3	YES
0	0	1	0	12	4	YES
0	0	1	1	20	7	YES
0	1	0	0	22	7	YES
0	1	0	1	30	10	YES
0	1	1	0	34	11	YES
0	1	1	1	42	14	YES
1	0	0	0	16	5	YES
1	0	0	1	24	8	YES
1	0	1	0	28	9	YES
1	0	1	1	36	12	YES
1	1	0	0	38	12	YES
1	1	0	1	46	15	NO
1	1	1	0	50	16	NO
1	1	1	1	58	19	NO

The optimal solution is to build dams 2, 3 and 4. The budget is completely invested, and the expected revenues are 420 mCHF. Although the first dam has the highest return, it consumes too much of the budget. It is therefore better not to build it, and invest in the three others. It illustrates the fact that using the relaxed problem to make the decision yields to a sub-optimal solution.